2110201: COMP ENG MATH (2023/2) Quiz 6

Name: Student ID: Entry Number:

1. (4 points) Find an orthonormal basis that spans the same subspace as $\left\langle \begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}0\\6\\9\end{bmatrix}, \begin{bmatrix}4\\2\\0\end{bmatrix} \right\rangle$

   by using **Gram-Schmidt algorithm** only.

2. (4 points, 1 point each) Given a basis $B = \left\langle \begin{bmatrix}0\\0\\1/2\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}1/3\\0\\0\end{bmatrix} \right\rangle$

2.1. The matrix _____ maps vectors from the standard basis to vectors in the basis B.

2.2. The matrix _____ maps vectors from the basis B to vectors in the standard basis.

2.3. A vector $\begin{bmatrix}-1\\1\\-2\end{bmatrix}$ in a standard basis is a vector _____ in the basis B.

2.4. Given a basis D $= \left\langle \begin{bmatrix}-2\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} \right\rangle$ and a transformation T represented by a matrix $\mathbf{A} = \begin{bmatrix}2 & -1\\3 & 0\\-1 & 5\end{bmatrix}$,

   write $\mathrm{Rep}_{D,B}(T)$ as a multiplication of 3 matrices. (No need to compute the actual value.)

3. (2 points, 1 point each) Answer the following questions with a briefly explanation.

3.1. Given a transformation S: $\mathbf{R}^2 \Rightarrow \mathbf{R}^2$, given by $S(x, y) = \frac{1}{2}(x + y, x + y)$. Is S distortion-free?

3.2. What is **linear interpolation**? Can it be described using knowledges from **the change of basis**?